# BIG MODULAR — A0 / REVIEW ONLY

Complete pin-connected prototype circuit; NOT a validated power supply.
Default OFF. Dedicated 5 V or PD 9 V source only. No host-current negotiation.
Tentative simultaneous loads: 5 V input: 20 mA EACH rail; PD 9 V: 40 mA EACH rail.
No output ratings are warranted. User target currents, transient loads and noise limits remain unspecified.
PCB files: USB connector + mechanical holes ONLY. Absolutely NO tracks, vias or zones.
Read ../README.md before using or updating a PCB from schematic.

USB-C sink / permission / protected input

File: 01_input.kicad_sch

4.016 V intermediate supply / startup

File: 02_intermediate.kicad_sch

Bipolar preregulator: approximately +13.95 / −13.95 V

File: 03_bipolar.kicad_sch

6.52 V preregulator for clean +5 V

File: 04_five_prereg.kicad_sch

Low-noise postregulators / three output rails

File: 05_clean_rails.kicad_sch

16-pin Eurorack POWER OUT / harness only

File: 06_output.kicad_sch

NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

**Big Modular**

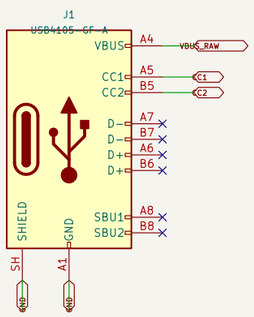
Sheet: /
File: big_modular.kicad_sch

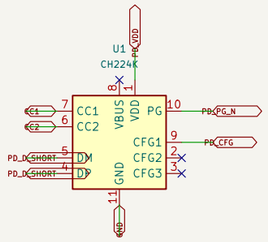
***Title: Big Modular USB-C PD to +12V / −12V / +5V***

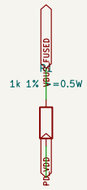
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|---|---|---|
| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
| KiCad E.D.A. 10.0.6 | | Id: 1/7 |

# USB-C sink / permission / protected input

9 V request only. No USB data. CH224K Rd must be verified on unpowered attach.
JP1 OPEN = OFF (default). Close ONLY for an approved dedicated source. PG is diagnostic, not a current-contract detector.
No series diode: TPS26600 reverse-current blocking is specified; qualify dead-battery/backfeed on hardware.
All bulk capacitance is behind the eFuse. No legacy-host enumeration or CC-current decoding is implemented.

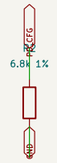


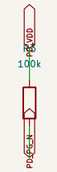



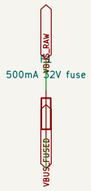



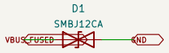


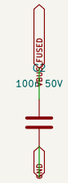


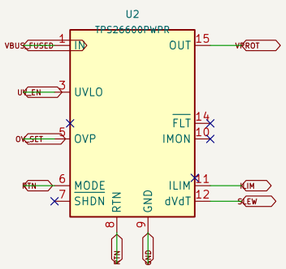


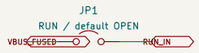



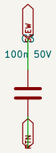



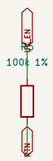



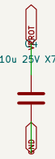


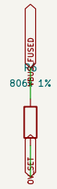


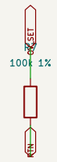



NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

**Big Modular**

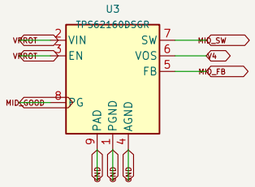
Sheet: /USB-C sink / permission / protected input/
File: 01_input.kicad_sch

***Title: USB-C sink / permission / protected input***

| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/7 |

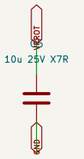
# 4.016 V intermediate supply / startup

TPS62160: 3..17 V input, nominal 2.25 MHz DCS control, internal compensation.
2.2uH / 22uF LC follows datasheet baseline. Require effective output C within datasheet stability range.
PG releases converter enables only after intermediate rail rises. PWM frequency is not an audio-noise guarantee.



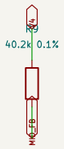


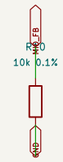


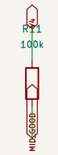


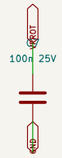







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**Big Modular**

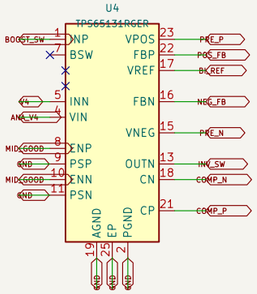
Sheet: /4.016 V intermediate supply / startup/
File: 02_intermediate.kicad_sch

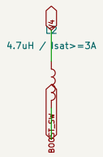
***Title: 4.016 V intermediate supply / startup***

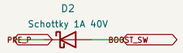
| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/7 |

# Bipolar preregulator: approximately +13.95 / −13.95 V

TPS65131: 2.7..5.5 V input; nominal 1.25 MHz. PSP/PSN LOW disables power-save mode.
Negative feedback references VREF, NOT ground. Positive/negative rectifier polarity follows boost/inverter energy paths.
BSW unused: positive preregulator can prebias through L/diode. No synchronized rail tracking promised.









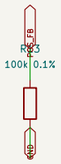


















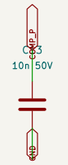






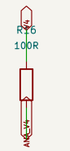


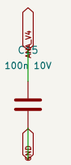


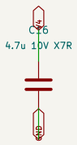












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**Big Modular**

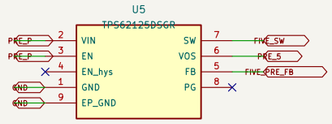
Sheet: /Bipolar preregulator: approximately +13.95 / −13.95 V/
File: 03_bipolar.kicad_sch

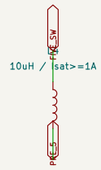
***Title: Bipolar preregulator: approximately +13.95 / −13.95 V***

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|---|---|---|
| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
| KiCad E.D.A. 10.0.6 | | Id: 4/7 |

# 6.52 V preregulator for clean +5 V

TPS62125 is explicitly used: TPS62160 cannot be substituted here (6 V recommended output limit).
10uH / 22uF combination is datasheet Table 2 recommendation for 5..10 V output.
DCS control operates UP TO 1 MHz and enters PFM at light load: strict always-MHz requirement is NOT met.

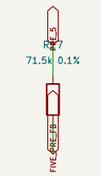


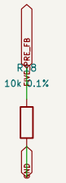



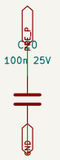







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**Big Modular**

Sheet: /6.52 V preregulator for clean +5 V/
File: 04_five_prereg.kicad_sch

***Title: 6.52 V preregulator for clean +5 V***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
| KiCad E.D.A. 10.0.6 | | Id: 5/7 |

# Low-noise postregulators / three output rails

Positive LDO VFB=1.185 V; negative LDO VFB=-1.176 V (not the headline reference values).
Negative IN and EN connect to PRE_N; GND and exposed pad stay at 0 V. All capacitors here are non-polar MLCC.
10uF outputs must retain >=2.2uF under bias. 10nF NR/SS + feedforward: verify overshoot and startup on bench.
Diodes bypass reverse differential and clamp wrong-polarity excursions, NOT precise output overvoltage protection.

U6
TPS7A4901DGNR
PRE_P 8 IN
PRE_P 5 EN
P12_NR 6 NR/SS
3 NC
7 DNC
OUT 1 +12
FB 2 P12_FB
GND 4 GND
EP_GND 9 GND

C21
10u 25V X7R
+12
GND

C22
10u 25V X7R
+12
GND

R30
91.2k 0.1%
+12
P12_FB

R20
10k 0.1%
P12_FB
GND

C23
10n 50V
P12_NR
GND

C24
10n 50V
+12
P12_FB

D4
Schottky 1A 40V
PRE_P
+12

D5
Schottky 1A 40V
+12
GND

U7
TPS7A3001DGNR
PRE_N 8 IN
PRE_N 5 EN
N12_NR 6 NR/SS
3 NC
7 DNC
OUT 1 -12
FB 2 N12_FB
GND 4 GND
EP_GND 9 GND

C25
10u 25V X7R
PRE_N
GND

C26
10u 25V X7R
-12
GND

R31
92k 0.1%
-12
N12_FB

R22
10k 0.1%
N12_FB
GND

C27
10n 50V
N12_NR
GND

C28
10n 50V
-12
N12_FB

D6
Schottky 1A 40V
+12
PRE_N

D7
Schottky 1A 40V
GND
+12

U8
TPS7A4901DGNR
PRE_5 8 IN
PRE_5 5 EN
P5_NR 6 NR/SS
3 NC
7 DNC
OUT 1 +5
FB 2 P5_FB
GND 4 GND
EP_GND 9 GND

C29
10u 25V X7R
+5
GND

C30
10u 25V X7R
+5
GND

R33
32.2k 0.1%
+5
P5_FB

R24
10k 0.1%
P5_FB
GND

C31
10n 50V
P5_NR
GND

C32
10n 50V
+5
P5_FB

D8
Schottky 1A 40V
PRE_5
+5

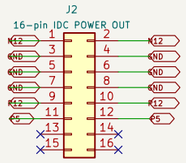
D9
Schottky 1A 40V
+5
GND

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**Big Modular**

| | | |
|---|---|---|
| Sheet: /Low-noise postregulators / three output rails/ | | |
| File: 05_clean_rails.kicad_sch | | |
| ***Title: Low-noise postregulators / three output rails*** | | |
| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
| KiCad E.D.A. 10.0.6 | | Id: 6/7 |

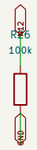
# 16-pin Eurorack POWER OUT / harness only

+5 V is on a separate STANDARD 16-pin interface, never added to the 10-pin standard.
J2 is an OFF-BOARD harness connector, not placed on either PCB. Exact crimp/IDC part and strain relief remain open.
RED STRIPE / pins 1 and 2 = −12 V. CV (13/14) and GATE (15/16) intentionally NC.
Do not trust key position alone. Continuity-test each cable; do not parallel this output with another supply.








































NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

**Big Modular**

Sheet: /16-pin Eurorack POWER OUT / harness only/
File: 06_output.kicad_sch

***Title: 16-pin Eurorack POWER OUT / harness only***

| Size: A3 | Date: 2026-09-24 | **Rev: A0 ENGINEERING PROTOT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/7 |